

Indian Institute of Engineering Science and Technology, Shibpur
CST Third Semester Midterm Examination, September 2024
Signals and Systems (CS-2104)

There are seven questions of six marks each.
Each question has two related parts with equal weightage.
Answer any five questions OR ten parts.

Full Marks: 30

Time: 120 minutes

1. (a) Briefly explain the concept of linear time invariant systems.
(b) Describe how the impulse response of such system can be obtained.
2. (a) What is the connection between the frequency variables used in Fourier transform, Laplace transform and Z transform?
(b) Explain the mapping of the stable region of complex s plane to the stable region of z domain.
3. A second order system is described as $\ddot{y}(t) + 5\dot{y}(t) + 6y(t) = 3x(t)$.
(a) Obtain the transfer function of the system.
(b) What is the response of the system for step input?
4. A second order discrete system is described by the difference equation:
 $y[n] - 0.5y[n-1] + 0.04y[n-2] = 3x[n]$.
(a) What are the poles of the system in z domain?
(b) Represent the system using three different cascaded block diagrams.
5. A periodic square wave signal with peak to peak amplitude of 2 volts and frequency 10 KHz is passed through an integrator that has an integration coefficient of $0.01(ms)^{-1}$.
(a) Sketch the output signal waveform in time domain for one cycle.
(b) Show how this integrator can be realized in continuous domain.
6. Provide schemes for the following with underlying mathematics:
(a) Obtaining initial value, final value and all values of a discrete sequence from the z transform
(b) Obtaining the frequency components of a square wave with very large time period.
7. Write short notes on the following:
(a) Energy content of a periodic signal (discrete and continuous) in time domain and frequency domain.
(b) Sampling of a continuous signal and its effects in frequency domain.